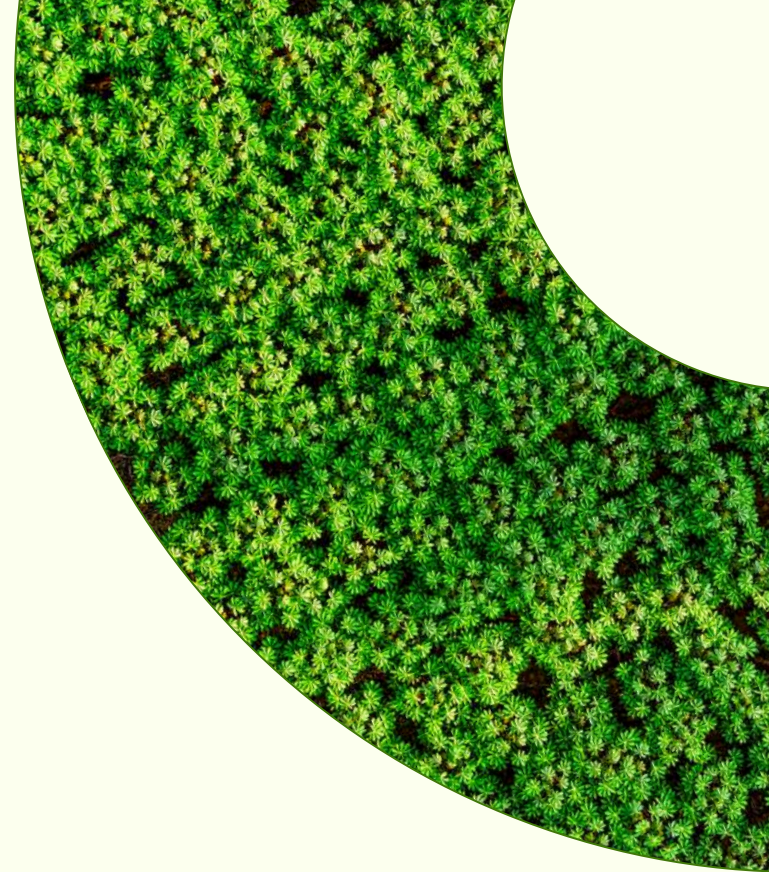




GNSS

AGRICULTURE DOMAIN



What is Precise Agriculture?

PRECISION AGRICULTURE (ALSO CALLED PRECISE AGRICULTURE) IS A MODERN FARMING APPROACH THAT USES ADVANCED TECHNOLOGIES—LIKE GPS, SENSORS, DRONES, AND DATA ANALYTICS—TO OPTIMIZE CROP AND LIVESTOCK PRODUCTION FOR HIGHER EFFICIENCY, SUSTAINABILITY, AND PROFITABILITY. IT FOCUSES ON MANAGING VARIABILITY WITHIN FIELDS, ENSURING THAT INPUTS SUCH AS WATER, FERTILIZER, AND PESTICIDES ARE APPLIED EXACTLY WHERE AND WHEN THEY ARE NEEDED



**THE BASIC FORM OF
AGRICULTURE STARTED JUST
10000 YEARS AGO**



CASE STUDIES WORKED ON -



GNSS IN INCREASING IRRIGATION



GNSS FOR BETTER ANALYSIS
OF SOIL HEALTH



OBJECTIVES

Sustainable irrigation for water deprived regions

How can we use GNSS to make irrigation smart

Irrigation and Rainfall correlation

We have shown how irrigation and rainfall are correlated



Smart Soil Health Monitoring iUsing GNSS
How can we introduce smart GNSS technologies to improve soil health

General Soil health factors and their impact
What are the effects of certain indicators of soil health

Country - India

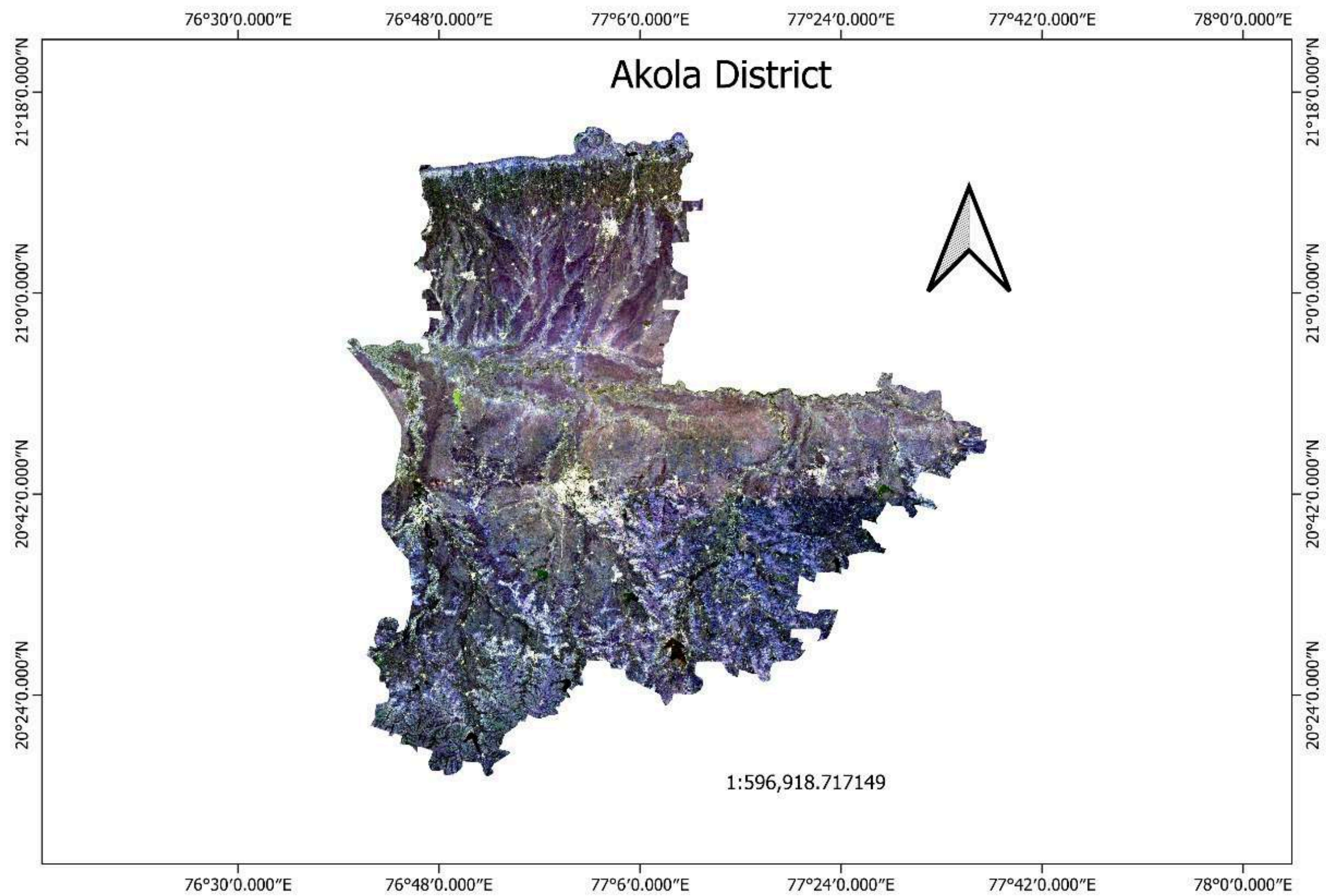


State - Maharashtra



District - Akola





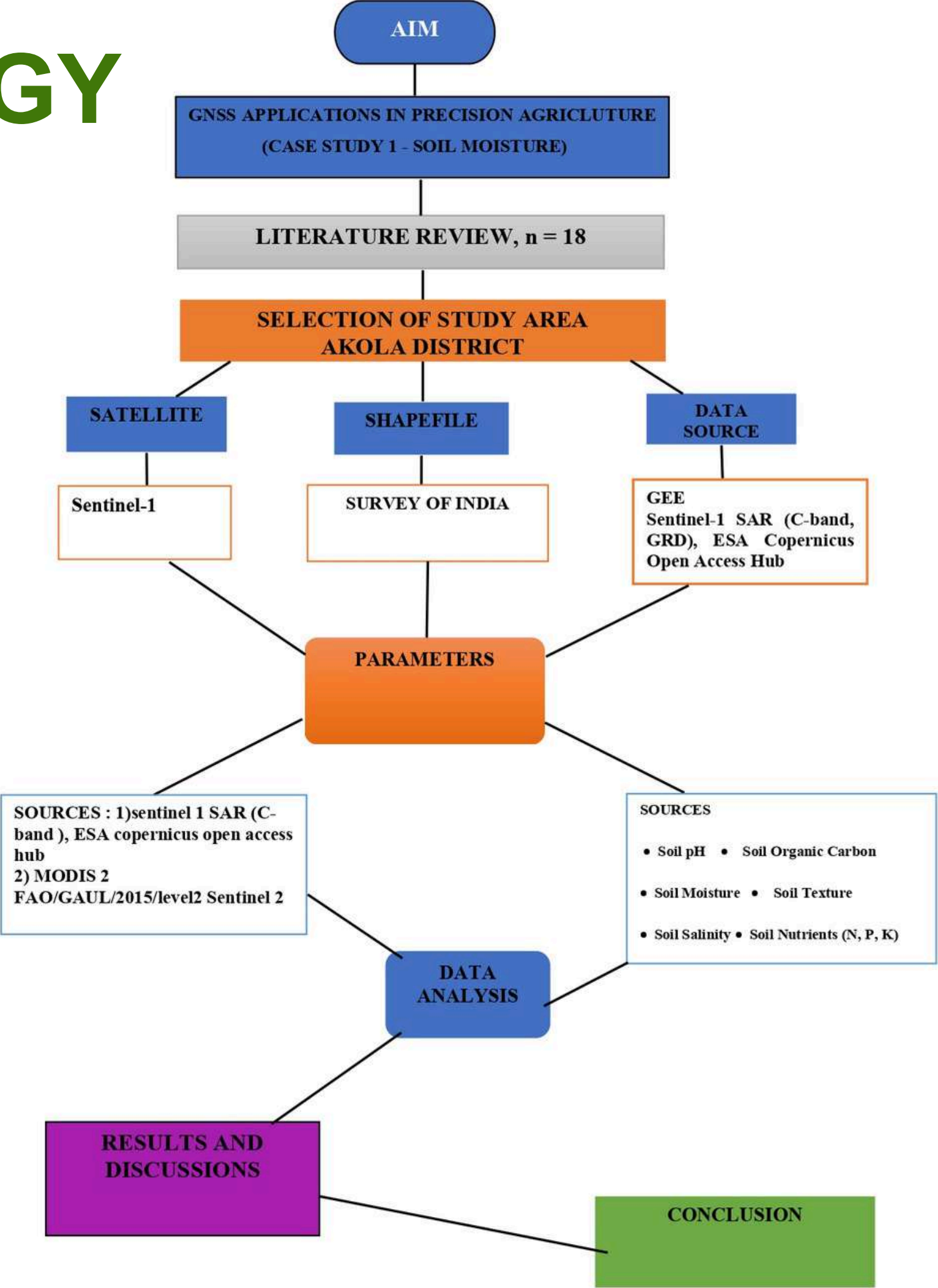


CASE STUDY 1

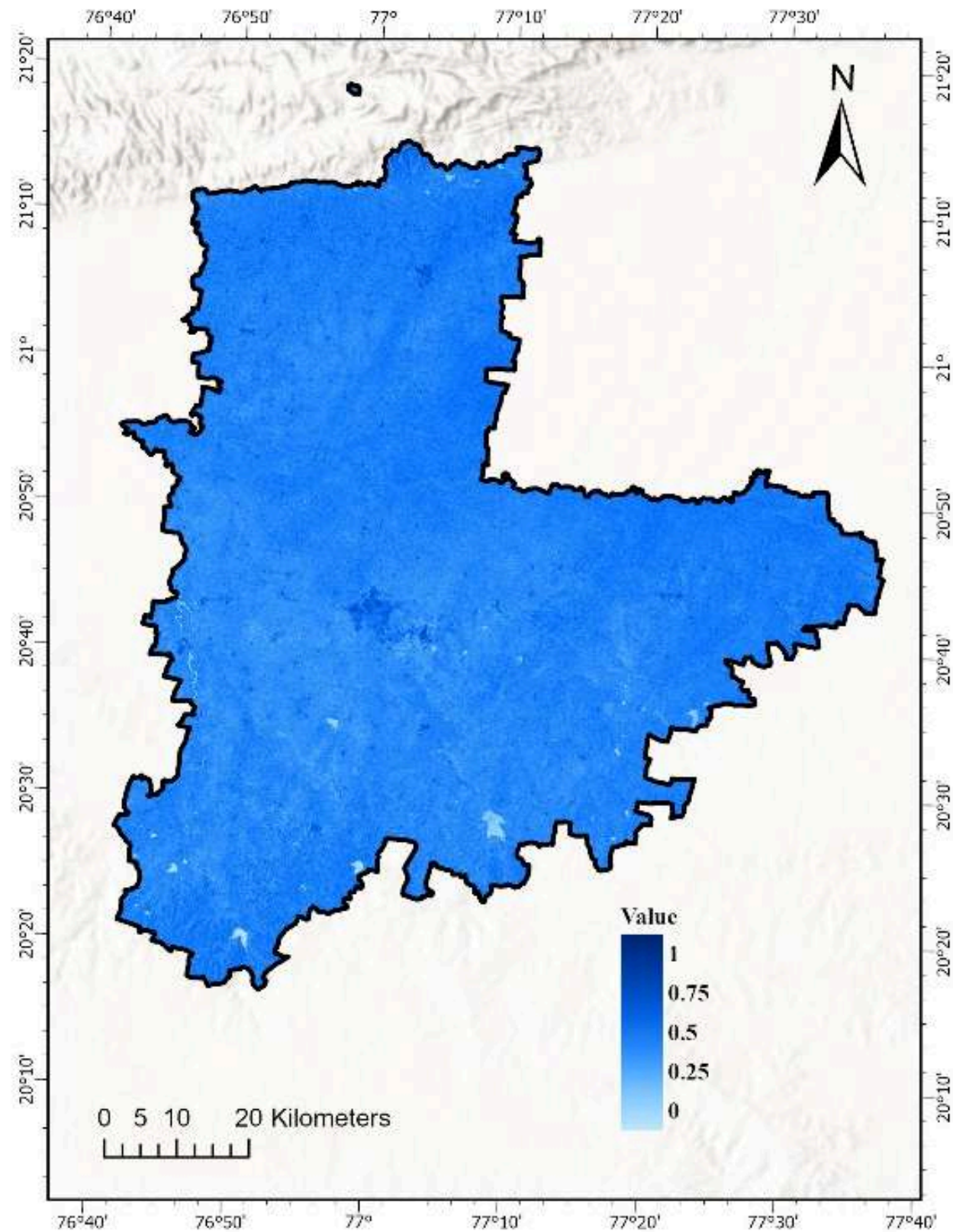
GNSS FOR BETTER ANALYSIS OF SOIL HEALTH



METHODOLOGY

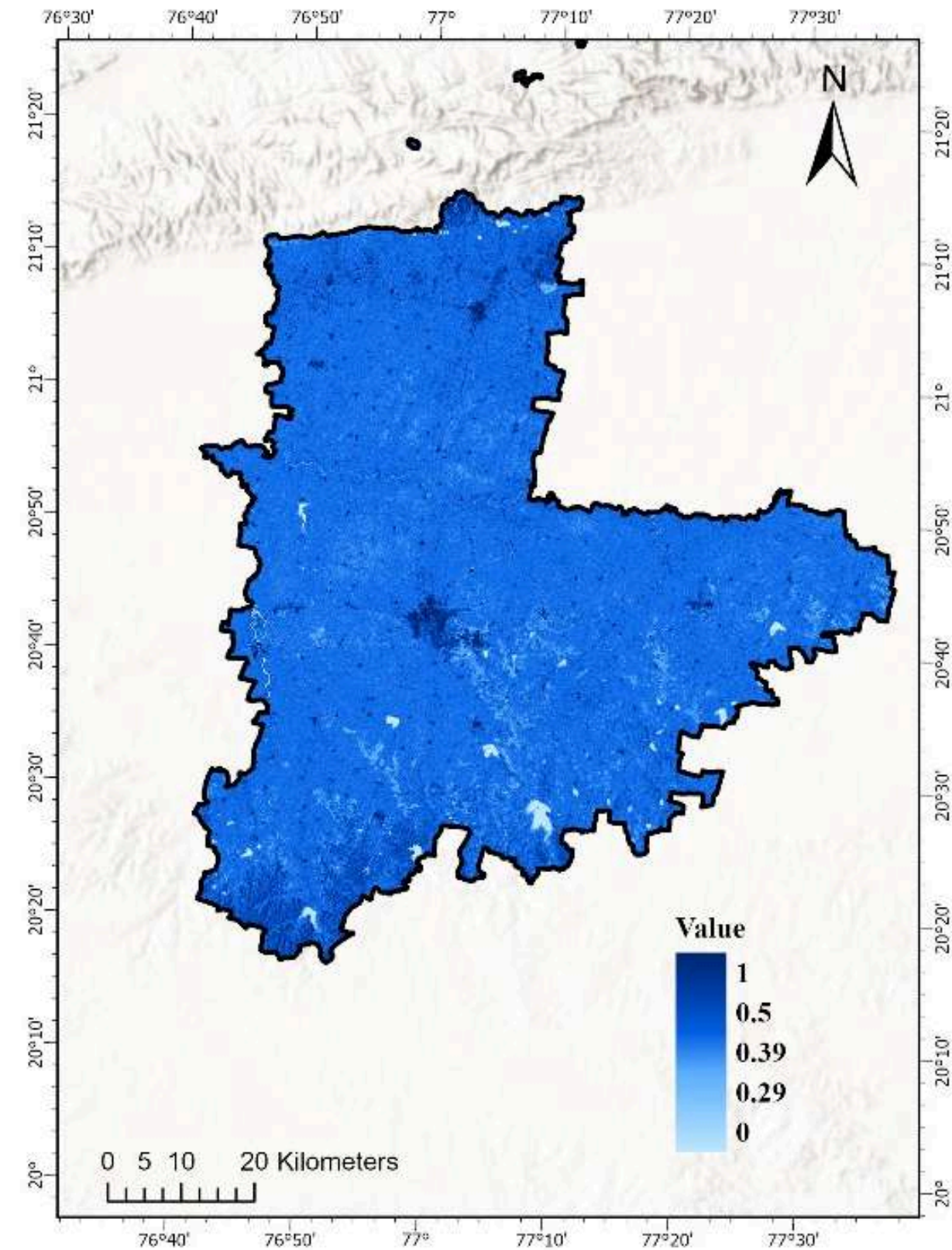


SOIL MOISTURE INDEX OF AKOLA 2015



Source: Sentinel-1 SAR (C-band), ESA / Copernicus Open Access Hub

SOIL MOISTURE INDEX OF AKOLA 2024



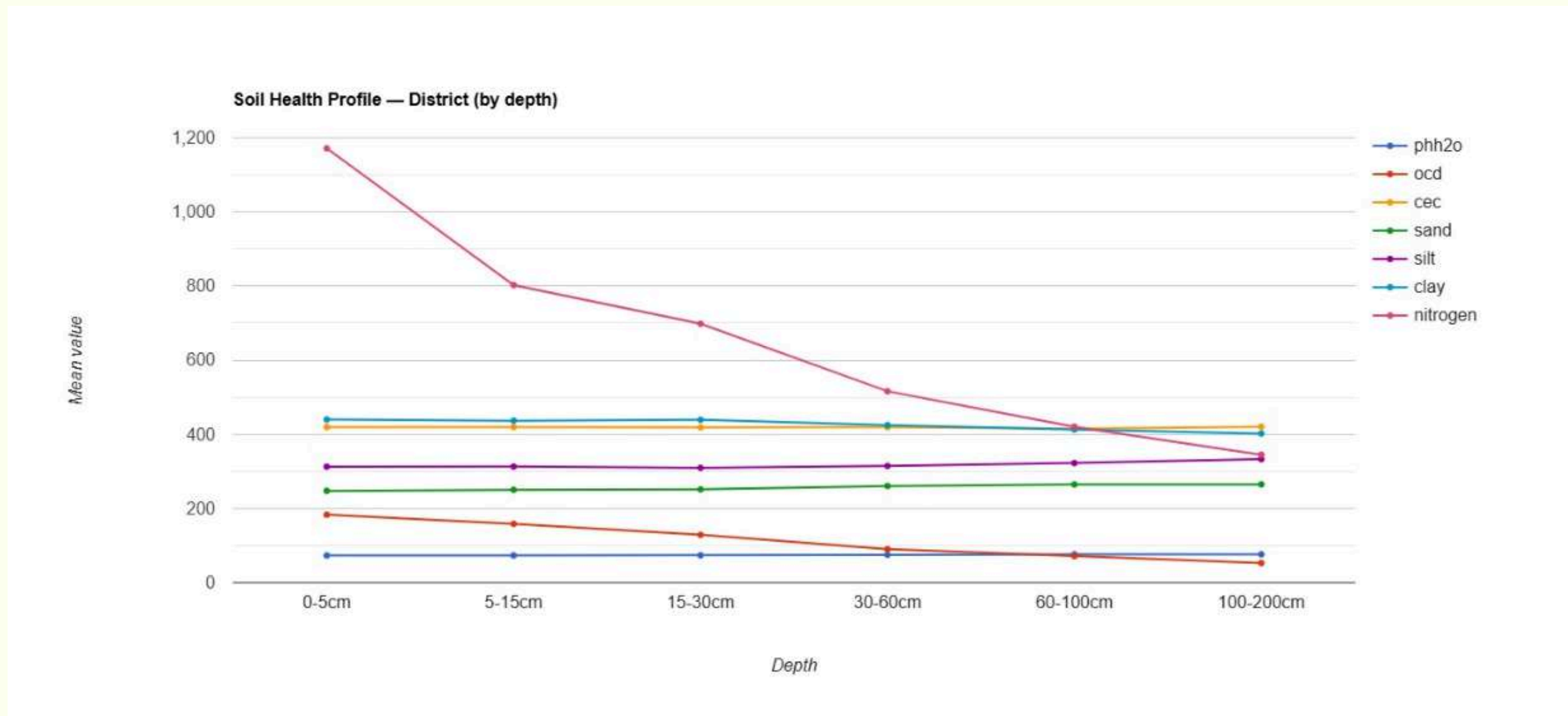
Source: Sentinel-1 SAR (C-band), ESA / Copernicus Open Access Hub

Interpretation

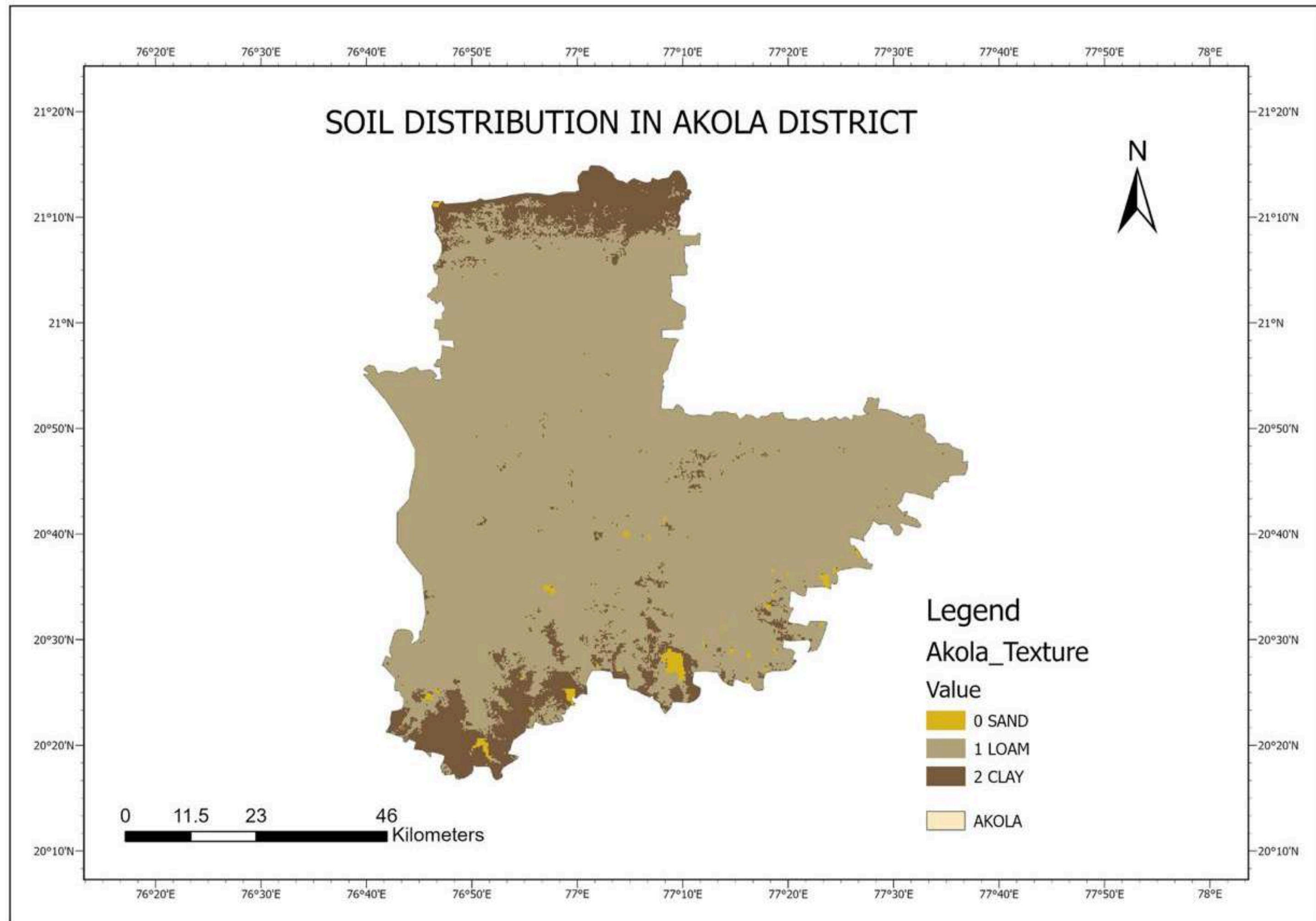
The two soil moisture maps from 2015 and 2024 give us a complete picture of moisture across Akola using satellite radar data. But the 2024 map reveals more detail. You can see small wet and dry patches that weren't visible at the district level before.

These maps are helpful but turning them into practical farming decisions requires more. You need accurate ground measurements and a system that can monitor and make decisions automatically.

That's where GNSS technology comes in. It becomes the foundation for making these maps useful. Let me walk you through how GNSS can help. I'll explain what each technique does, how to set it up based on the map, what accuracy you need, how many points to sample, and a step-by-step plan to make it work.



The soil health profile chart shows that the richest soil nutrients—especially nitrogen and organic carbon—are found in the topsoil and decrease rapidly as we go deeper. Meanwhile, soil texture (clay, sand, silt) and pH don't change much with depth. This means that plants get most of their nutrients from the top 0–30 cm of soil, while the deeper layers hold fewer nutrients but more moisture.



Interpretation

- 1. Clay-Dominated Areas (Dark Brown Zones — North & South Akola):** These areas have a lot of clay in the soil. This means the soil can hold water well, but at the same time also cracks and swells when it gets wet or dry, and sometimes loses water slowly. Even a small change in land level or moisture can make clay soils behave very differently, even within a short distance .
- 2. Soil Texture Transition Zones (Patchy Yellow/Brown Areas — Central & South-East Akola):** These are spots where the soil type changes a lot, from sand to loam to clay. On the map, this looks like a mix of yellow and brown patches. As the soil keeps changing, some areas might be rich in nutrients and others might not, even if they lie very close together.
- 3. Sandy Pockets (Yellow Zones — North-east & South-central Akola):** Even though these sandy areas are small, they still matter. Sandy soil doesn't hold onto water or nutrients very well, so it dries out quickly and plants can struggle here. It's a good idea to check the soil moisture often in these spots, using sensors that record their exact location with GNSS. This way, you can keep a close watch on dry areas and get better advice for crops that might be affected by drought.

GNSS ENABLED SOLUTIONS

Problem 1: Varying soil moisture leads to wastage of irrigation and results in overall poor management of soil

Solution: Equipping farmers with GNSS enabled soil sensors placed at fixed locations in the field. Each sensor sends exact location + moisture reading.

Problem 2: Over-irrigation in some areas

GNSS marks exact boundary of irrigation zones. Automatic valves run only in zones with low moisture

Problem 3: Soil nutrient loss due to excess irrigation

Tractors with GNSS apply fertilizer only in areas showing low soil health. Avoids applying fertilizer in already wet zones where nutrients would wash away.

PROBLEM 4: FARMERS DON'T KNOW WHERE SOIL IS DEGRADING

Collect soil samples from GNSS-marked spots every year.
Compare soil health over time at the exact same locations

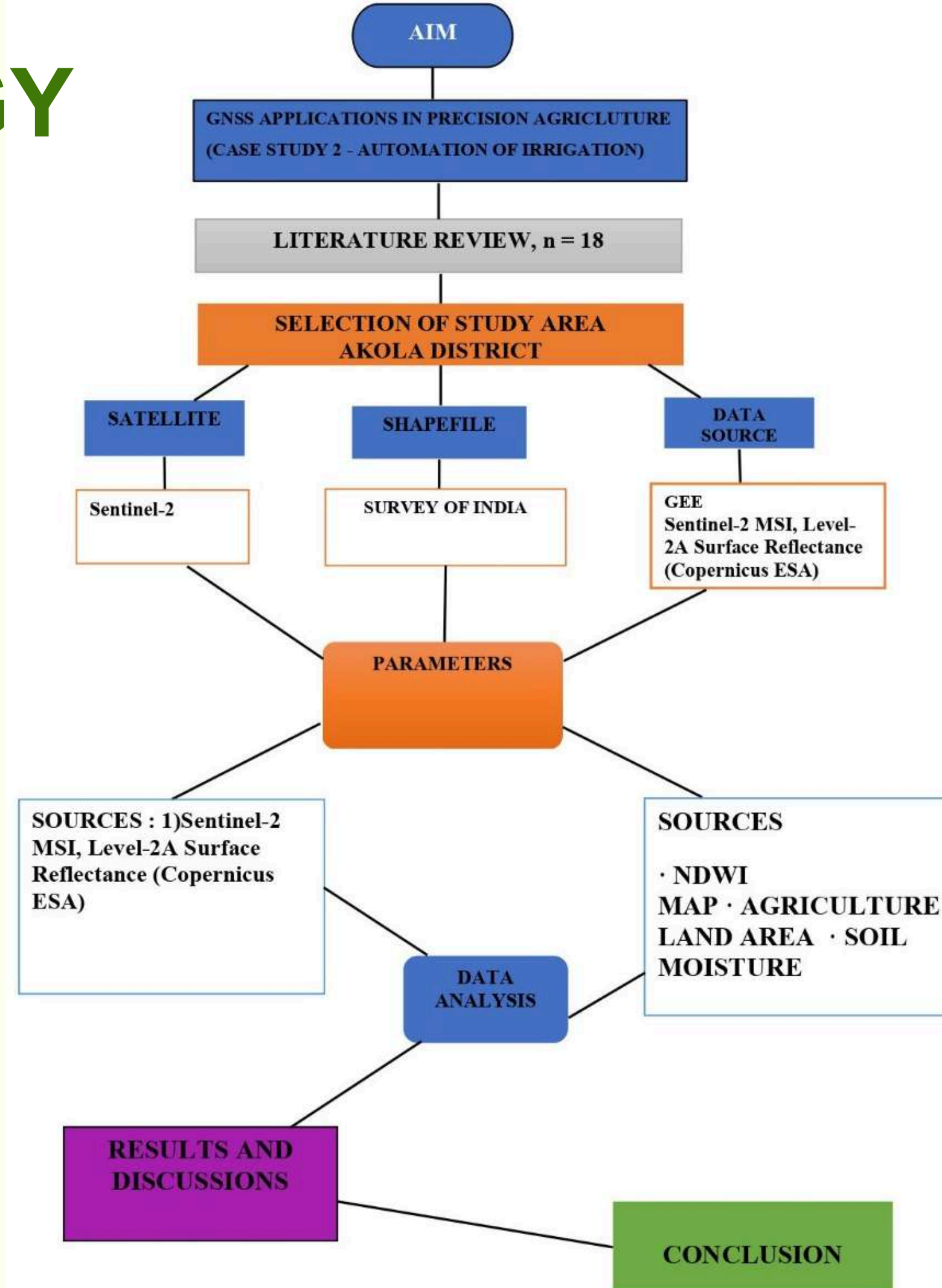


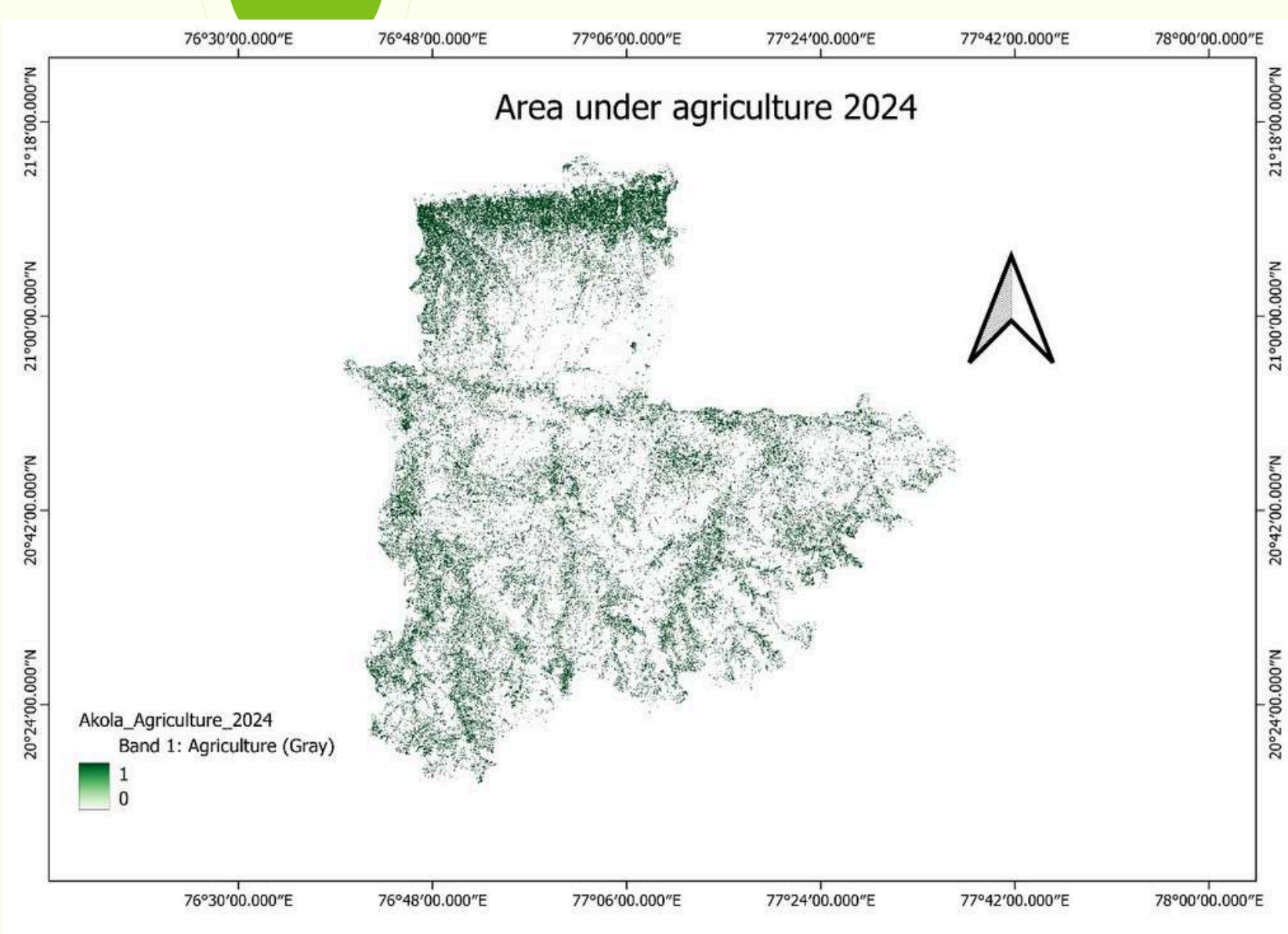


CASE STUDY 2

IRRIGATION AUTOMATION IN AKOLA

METHODOLOGY



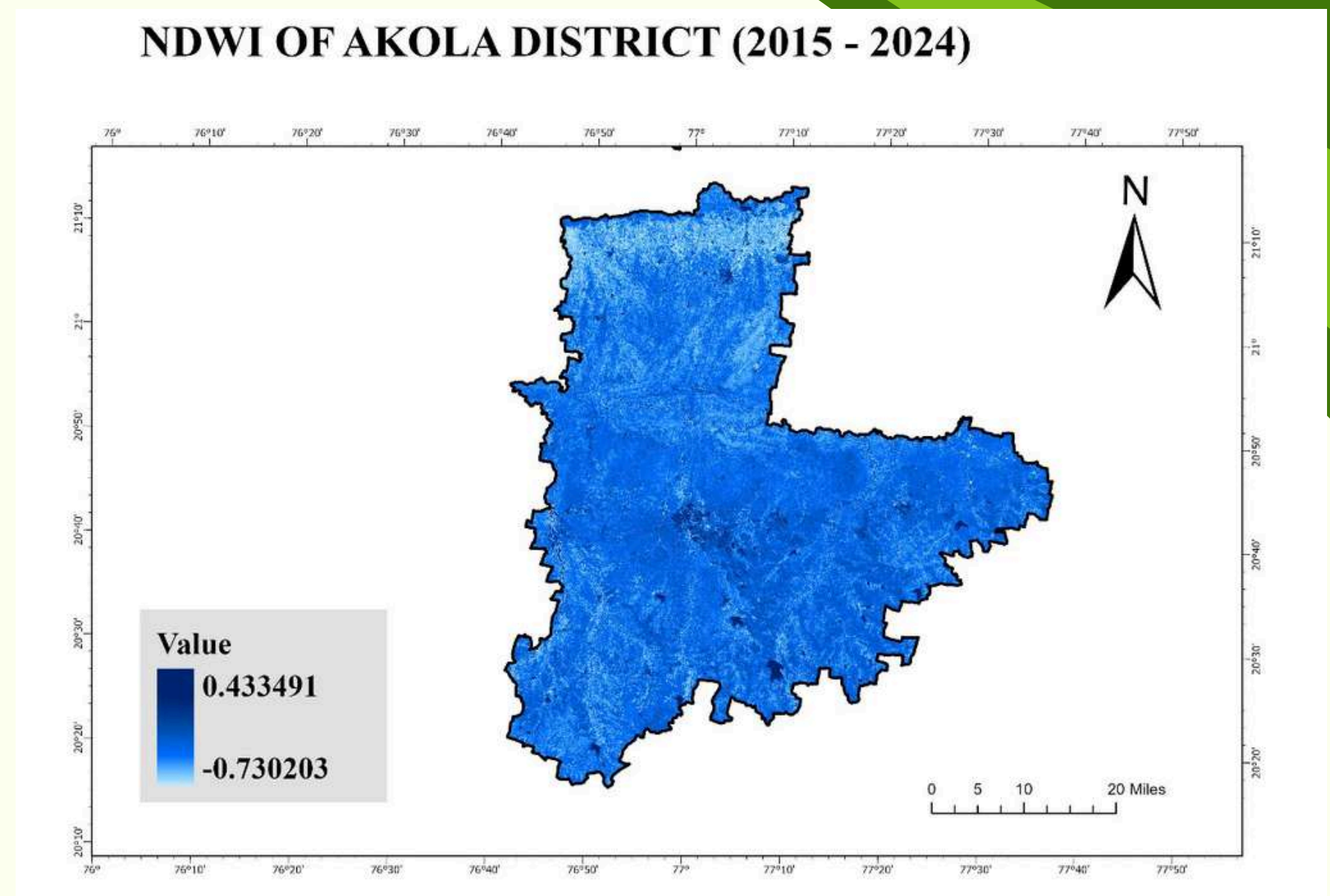


AGRICULTURAL AREA SPATIAL DISTRIBUTION

The agricultural area map of Akola district clearly delineates spatial distribution of agricultural fields.

The northern part of the district forms the core agricultural belt. Central and southern parts have fragmented agricultural areas. This shows areas

Northern part is where intense agricultural activities are done.

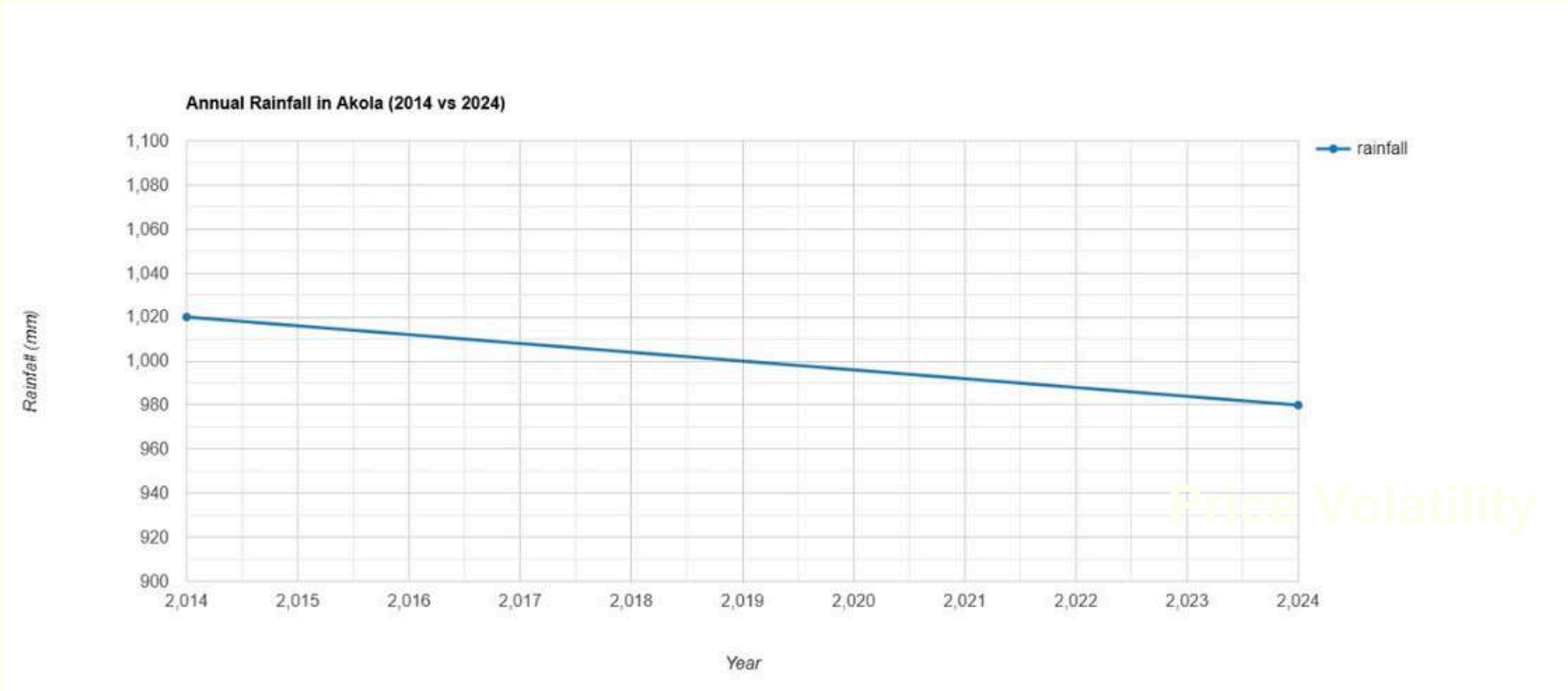


NDWI SPATIAL VARIATION

Values range from 0.433 to -0.73. Value ranges indicate low to moderately low surface water availability, indicating a semi-arid status across the district. Northern parts of the agriculture areas are having low NDVI values, This clearly shows the reliance of rainfall and irrigation for cultivation in the district. **This points towards the importance of irrigation in the district.**

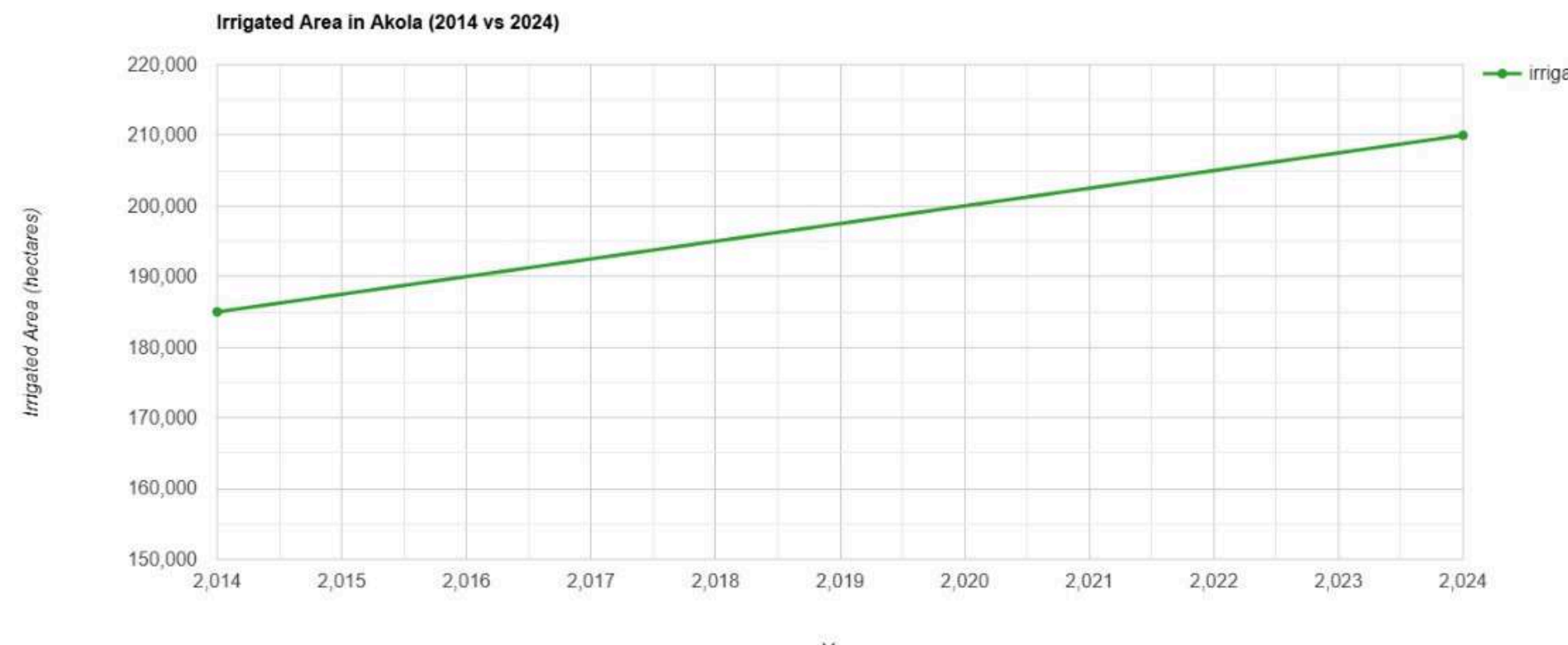


Annual Rainfall (2014-2024)



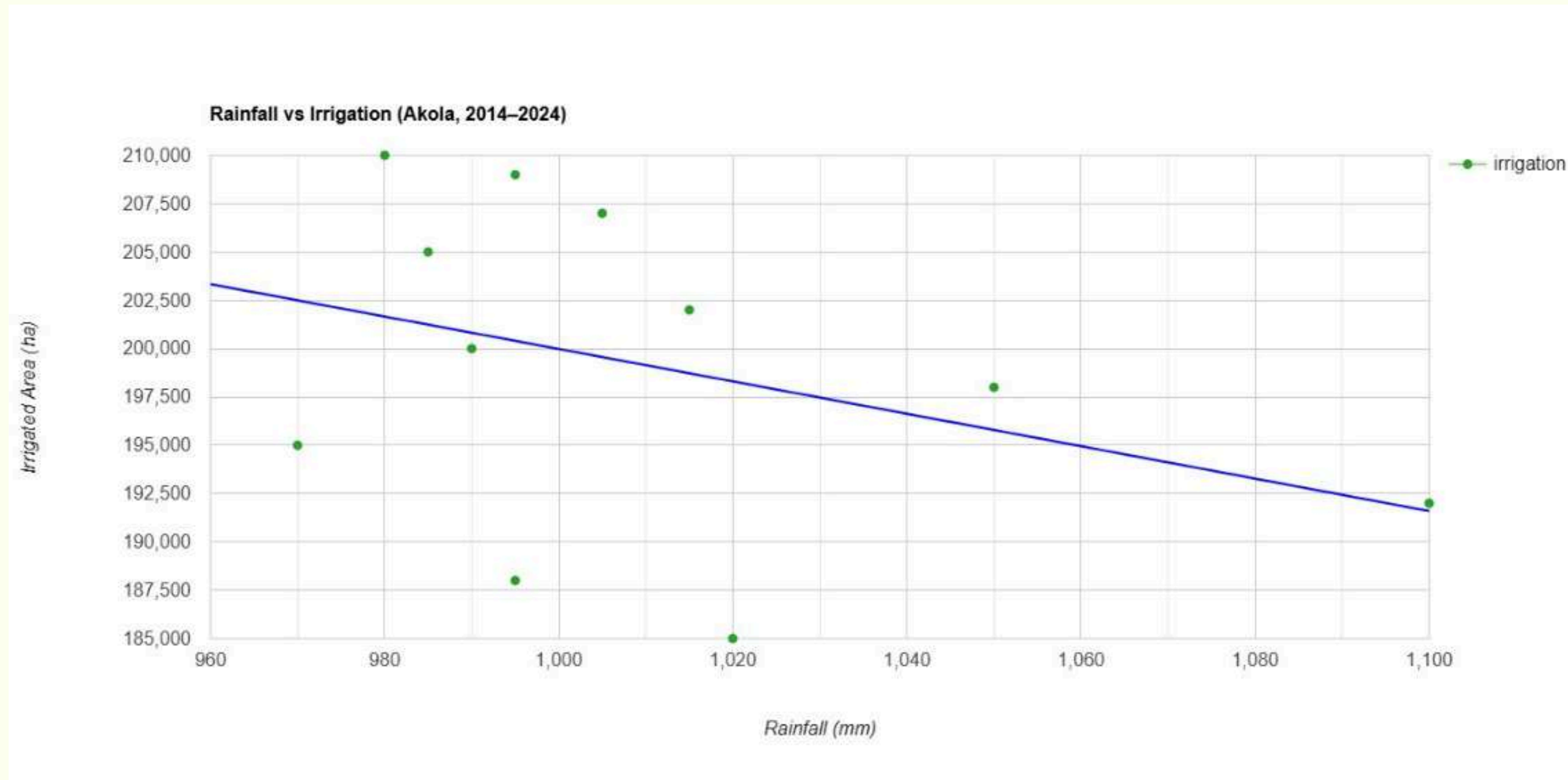
Declining trend of annual rainfall. In 2014 it was 1020mm it became around 980mm by 2024.

Irrigated Area(Ha) (2014-2024)



Ten years ago, the irrigated area in the district was around 180 thousand hectares; it increased to 210 thousand hectares. The increase is 16%

Rainfall Vs. Irrigation



As rainfall decreased in the district, agricultural areas relied more on irrigation to maintain water requirements. Even though there is a negative correlation, it is only a weak association. Rainfall reduction in the district, and increase in irrigated area in the district was not proportional.

GNSS ENABLED INTERVENTIONS

IRRIGATION EFFICIENCY CAN BE INCREASED DRASTICALLY WITH GNSS INTEGRATION



Accurate field Boundary delineation

- Can be done by high precision by RTK or multi- GNSS receivers
- To map exact farm boundaries and farm infrastructure like wells, canals, ponds etc
- This results in better **targeted intervention, clear identification of under-irrigated areas in the district**

Integration with remote sensing data (NDWI/NDVI)

- Fuse GNSS ground truth with satellite NDWI/NDVI to monitor moisture, identify stress pockets, and trigger alerts for targeted irrigation. Can build simple dashboards
- Improves accuracy drastically
- **Identification of stress hotspots improves irrigation efficiency.**

Precision land levelling

- Using GNSS guide laser graders to level the field,
- Will result in uniform infiltration of irrigated water, so that less water is needed per hectare. Leads to irrigation efficiency and uniform crop productivity highly relevant for arid regions.

GNSS-assisted irrigation scheduling

- Tag soil moisture sensors, on-farm ponds and canals with GNSS coordinates and use location-aware irrigation controllers or farmer apps. For mechanized irrigation, use GNSS guidance for swath/pivot control.
- Better timing and spatial targeting - less water demand overall.

Mapping and managing water sources

- Map ponds, canals, borewell locations, recharge structures, and impervious urban runoff sinks, monitor their change with GNSS-tagged repeat surveys.
- Lead to increased water security without increasing irrigated footprint, by tracking water sources on real time basis , identifying potential for storing monsoon water etc.,

CONCLUSION

1. Soil texture controls irrigation need.
 - Clay-rich southern Akola retains high moisture → less irrigation demand.
 - Loamy and sandy regions in central and southeastern Akola lose moisture quickly → high irrigation need.
2. Soil moisture has declined from 2015 to 2024
3. Irrigation is concentrated in eastern and southeastern zones, where soils are weaker in moisture retention.
4. Water availability (NDWI) is lowest where irrigation demand is highest
5. GNSS can significantly improve irrigation efficiency. Precision land leveling with GNSS ensures uniform water infiltration and reduces wastage.
6. GNSS-tagged soil moisture sensors enable exact irrigation timing and location-based scheduling.
7. GNSS-guided mechanized irrigation (pivot/strip systems) applies water only where required.
8. GNSS improves targeting of under-irrigated zones
9. GNSS-enabled precision agriculture can maintain yields while reducing water use and focusing irrigation where it gives the most return. Helps identify areas with poor water access and plan new water structures.

Thank You!

